Nabeeha Mahanta

Contact Information

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Professional Summary

Analytical and detail-oriented MSc Advanced Astronomy and Astrophysics student with a strong foundation in data analysis, computational modelling, research, and scientific communication. Experienced in working with complex scientific datasets, using Python for analysis and visualisation, and interpreting technical results with accuracy. Skilled in independent research, problem-solving, collaboration, mentoring, and presenting complex information clearly to both technical and non-technical audiences. Adaptable and motivated to apply strong quantitative, analytical, and communication skills across research, data, technology, consulting, project coordination, and graduate-level roles.

Work Experience

Data Analyst | Metroll
New South Wales, Australia | June 2026 – Present

- Analyse sales, inventory, order and invoice data across multiple branches.
- Built reports on sales representative and branch performance, and on how rebates affect results.
- Built a desktop application that converts invoices into Excel spreadsheets, replacing manual data entry.
- Clean and prepare data using SQL, Python and Excel, and present findings to technical and non-technical teams.

Research Assistant Internship
Aberystwyth, United Kingdom | January 2024 – January 2025

- Utilized GCS (Graduated Cylindrical Shell) modeling to analyze the propagation of Coronal Mass Ejections (CMEs), contributing to the department's solar physics research.
- Designed and maintained a comprehensive ICME (Interplanetary Coronal Mass Ejection) catalog, ensuring accuracy and consistency in data for future research.
- Collaborated with senior researchers to analyze solar wind patterns, identifying key indicators for space weather forecasting.
- Regularly presented complex data insights to the team, using visualization tools and simulations to effectively communicate results.
- Authored detailed research reports and documentation, enhancing the accessibility of data for ongoing and future studies.

Physics Mentorship | Cardiff University
Remote | October 2022 – March 2023

- Mentored students in physics principles, leading group sessions and one-on-one mentoring to improve their academic performance and foster a deeper understanding of concepts.
- Supervised multiple school-level physics projects, guiding students in designing experiments, collecting data, and presenting their findings.
- Developed tailored educational resources and lesson plans to support diverse student learning needs.
- Engaged in public outreach programs to promote interest in physics among young students across Wales, increasing enrollment in the mentoring program.
- Led workshops on key scientific topics, simplifying advanced physics concepts into digestible formats for younger audiences.

Education

MSc (Advanced) in Astronomy and Astrophysics | Australian National University
Canberra, Australia | Start: Feb 2026

- Building advanced knowledge in astrophysical theory, observational astronomy, stellar and galactic processes, and computational methods used in modern astronomical research.
- Developing strong analytical and research skills through coursework involving scientific problem-solving, data interpretation, modelling, and the presentation of complex astrophysical concepts.

BSc Astrophysics | Aberystwyth University
Aberystwyth, United Kingdom | June 2025

Awarded First Class Honours

Electricity and Magnetism, Quantum Mechanics, Mathematical Physics, Stars and Planets, Astrophysics 1: Physics of the Sun, Astrophysics 2: Galaxies, General Relativity and Cosmology, Particles, Quanta and Fields.

Data analysis, numerical simulation, theoretical modeling, programming (Python, Visual Basic for applications (VBA)), data handling.

International Baccalaureate Diploma Programme (IBDP) | Woodstock School
Mussoorie, India | May 2021

Subjects: Physics HL, Chemistry HL, Math AA HL

International General Certificate of Secondary Education (IGCSE) | Good Shepherd International School
Ooty, India | May 2019

Subjects: Physics, Chemistry, Math

Projects

A Gaia-Based Framework for Measuring Stellar Mass Functions in Milky Way Star Clusters
Australian National University, Canberra | MSc research project, in progress

My MSc research project develops a Gaia-based framework for measuring stellar mass functions in Milky Way star clusters, to test whether the mass function is universal. Using Gaia DR3 data, the framework corrects cluster star counts for observational incompleteness and unresolved binary stars.

Relationship Between Coronal Mass Ejection Observables Near the Sun and Measurements Near Earth
Aberystwyth, United Kingdom | April 2025

My dissertation explored the relationship between coronal mass ejection (CME) observables near the Sun and their measurements near Earth. I analyzed 292 CME–ICME pairs from Solar Cycles 23–25 using multi-catalogue data, GCS modeling, and statistical methods in Python. The study contributes to improving space weather forecasting by linking solar observations with geomagnetic impacts.

Skills

Research skills, teamwork, data analysis, data handling, research consulting, public speaking, presentation skills, project management, communication, problem solving, critical thinking, time management

References

Available upon request.